



Website Traffic Analysis

Objective

The objective of this project is to analyze website interaction data to understand user engagement patterns across different events, countries, cities, artists, albums, and tracks. The analysis focuses on identifying user activity trends, popular content, and geographic distribution to generate actionable business insights that can help Alfido Tech improve user engagement and optimize marketing strategies. The project includes data cleaning, exploratory data analysis (EDA), visualization, and business recommendations based on the available event-level interaction data.

Importing Libraries

```
In [1]: import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use("default")
```

```
In [2]: df=pd.read_csv("traffic.csv")
```

Dataset Overview

```
In [3]: df.head()
```

```
Out[3]:
```

	event	date	country	city	artist	album	track	isrc
0	click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741
1	click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741
2	click	2021-08-21	India	Ludhiana	Reyanna Maria	So Pretty	So Pretty	USUM72100871
3	click	2021-08-21	France	Unknown	Simone & Simaria, Sebastian Yatra	No Llores Más	No Llores Más	BRUM72003904
4	click	2021-08-21	Maldives	Malé	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741

```
In [4]: df.shape
```

```
Out[4]: (226278, 9)
```

```
In [5]: df.columns
```

```
Out[5]: Index(['event', 'date', 'country', 'city', 'artist', 'album', 'track', 'isrc',  
              'linkid'],  
              dtype='object')
```

```
In [6]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 226278 entries, 0 to 226277  
Data columns (total 9 columns):  
#   Column      Non-Null Count  Dtype  
---  -  
0   event       226278 non-null  object  
1   date        226278 non-null  object  
2   country     226267 non-null  object  
3   city        226267 non-null  object  
4   artist      226241 non-null  object  
5   album       226273 non-null  object  
6   track       226273 non-null  object  
7   isrc        219157 non-null  object  
8   linkid      226278 non-null  object  
dtypes: object(9)  
memory usage: 15.5+ MB
```

```
In [7]: df.describe(include='all')
```

```
Out[7]:
```

	event	date	country	city	artist	album	track	
count	226278	226278	226267	226267	226241	226273	226273	
unique	3	7	211	11993	2419	3254	3562	
top	pageview	2021-08-19	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2
freq	142015	35361	47334	22791	40841	40841	40841	

Data Cleaning

```
In [8]: #checking null values  
df.isnull().sum()
```

```
Out[8]: event      0
       date      0
       country    11
       city      11
       artist     37
       album      5
       track      5
       isrc      7121
       linkid     0
       dtype: int64
```

```
In [9]: #checking for duplicates
       df.duplicated().sum()
```

```
Out[9]: 103711
```

```
In [10]: #dropping duplicates
       df.drop_duplicates(inplace=True)
```

```
In [11]: df.duplicated().sum()
```

```
Out[11]: 0
```

```
In [12]: df.isnull().sum()
```

```
Out[12]: event      0
       date      0
       country      5
       city      5
       artist     28
       album      4
       track      4
       isrc     6306
       linkid     0
       dtype: int64
```

Replacing null values

```
In [13]: df['country'] = df['country'].fillna('Unknown')
```

```
In [14]: df['city'] = df['city'].fillna('Unknown')
```

```
In [15]: df['artist'] = df['artist'].fillna('Unknown Artist')
```

```
In [16]: df['album'] = df['album'].fillna('Unknown Album')
```

```
In [17]: df['track'] = df['track'].fillna('Unknown Track')
```

```
In [18]: df['isrc'] = df['isrc'].fillna('Unknown')
```

```
In [19]: df.isnull().sum()
```

```
Out[19]: event      0
         date      0
         country    0
         city      0
         artist    0
         album     0
         track     0
         isrc      0
         linkid    0
         dtype: int64
```

```
In [20]: df.duplicated().sum()
```

```
Out[20]: 0
```

```
In [21]: #date conversion
         df['date']=pd.to_datetime(df['date'])
```

Feature Engineering

```
In [22]: df['Month']=df['date'].dt.month_name()
```

```
In [23]: df['Year']=df['date'].dt.year
```

```
In [24]: df['Day']=df['date'].dt.day_name()
```

KPI Analysis

```
In [25]: #Total Events
         print("Total Events:",len(df))
```

Total Events: 122567

```
In [26]: #Total Unique Links
         print("Unique Links:",df['linkid'].nunique())
```

Unique Links: 3839

```
In [27]: #Total Countries
         print("Countries:",df['country'].nunique())
```

Countries: 212

```
In [28]: #Total Cities
         print("Cities:",df['city'].nunique())
```

Cities: 11993

```
In [29]: #Event Distribution
         print(df['event'].value_counts())
```

```
event
pageview    73360
click       32499
preview     16708
Name: count, dtype: int64
```

Visualization

In [30]: *#Event Distribution*

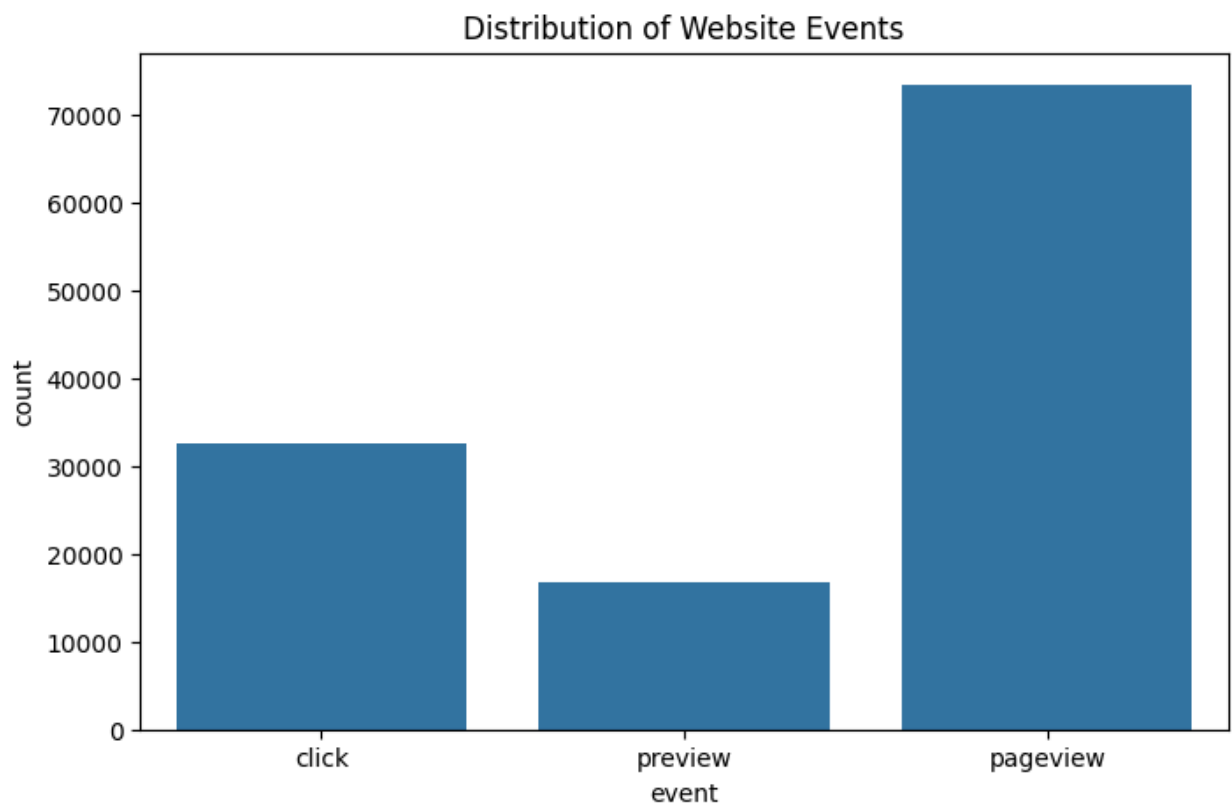
```
plt.figure(figsize=(8,5))

sns.countplot(data=df,x='event')

plt.title("Distribution of Website Events")

plt.savefig("events.png",dpi=300,bbox_inches="tight")

plt.show()
```



In [31]: *#Top Countries*

```
top_country=df['country'].value_counts().head(10)

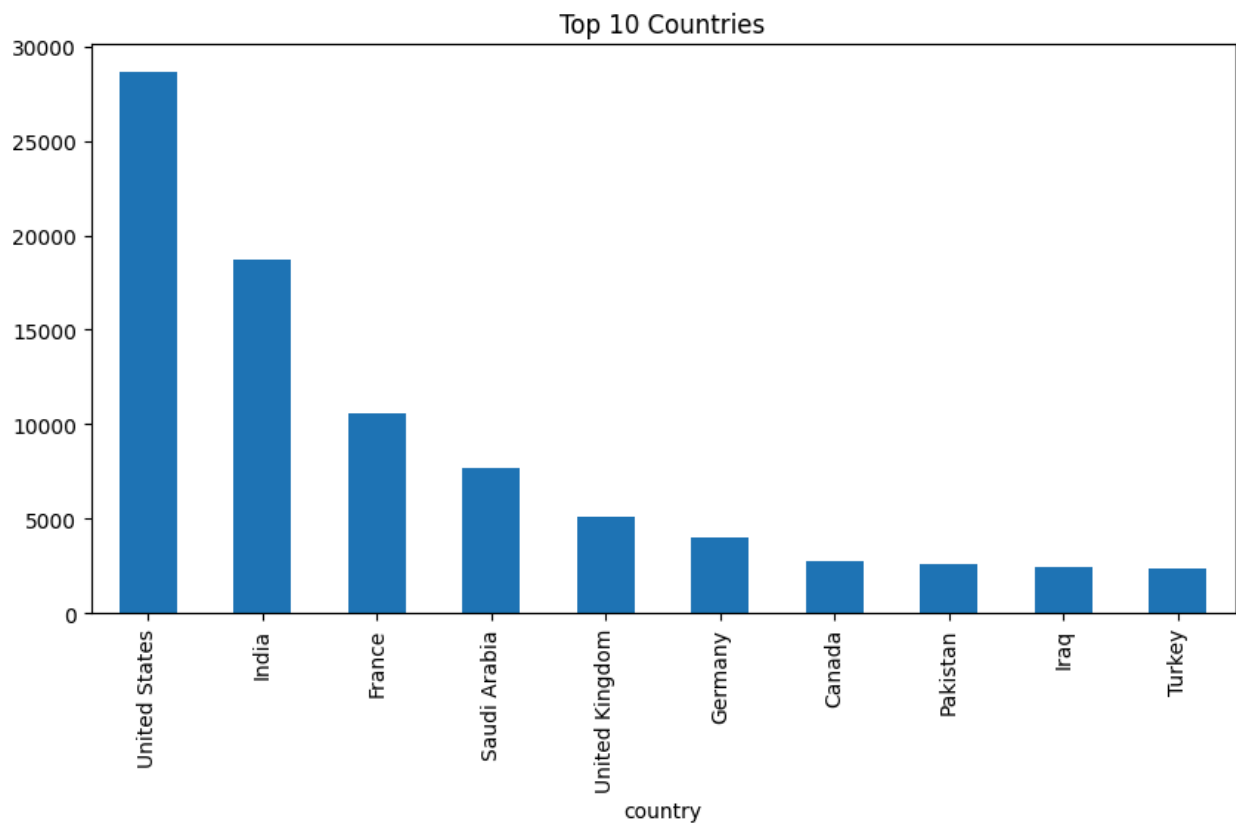
plt.figure(figsize=(10,5))

top_country.plot(kind='bar')
```

```
plt.title("Top 10 Countries")

plt.savefig("country.png",dpi=300,bbox_inches="tight")

plt.show()
```



```
In [32]: #Top Cities

top_city=df['city'].value_counts().head(10)

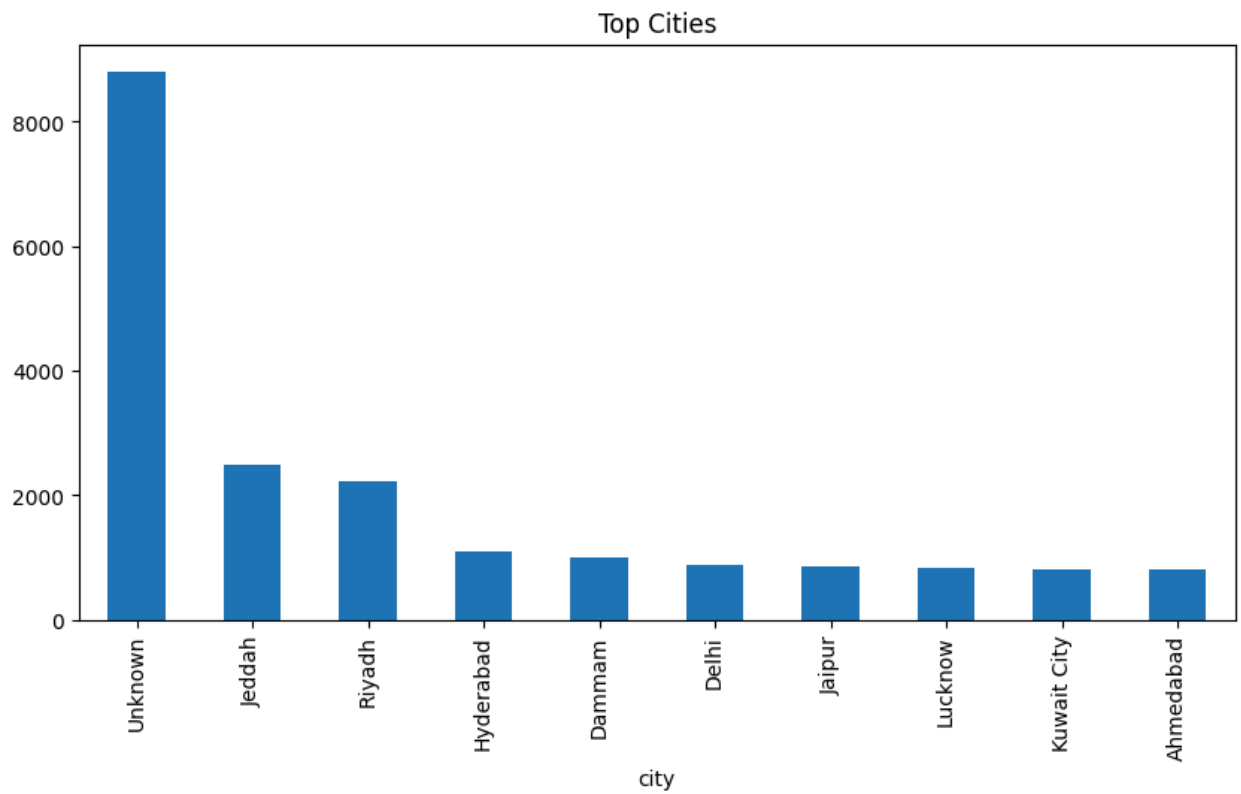
plt.figure(figsize=(10,5))

top_city.plot(kind='bar')

plt.title("Top Cities")

plt.savefig("cities.png",dpi=300,bbox_inches="tight")

plt.show()
```



```
In [33]: #Top Artists

artist=df['artist'].value_counts().head(10)

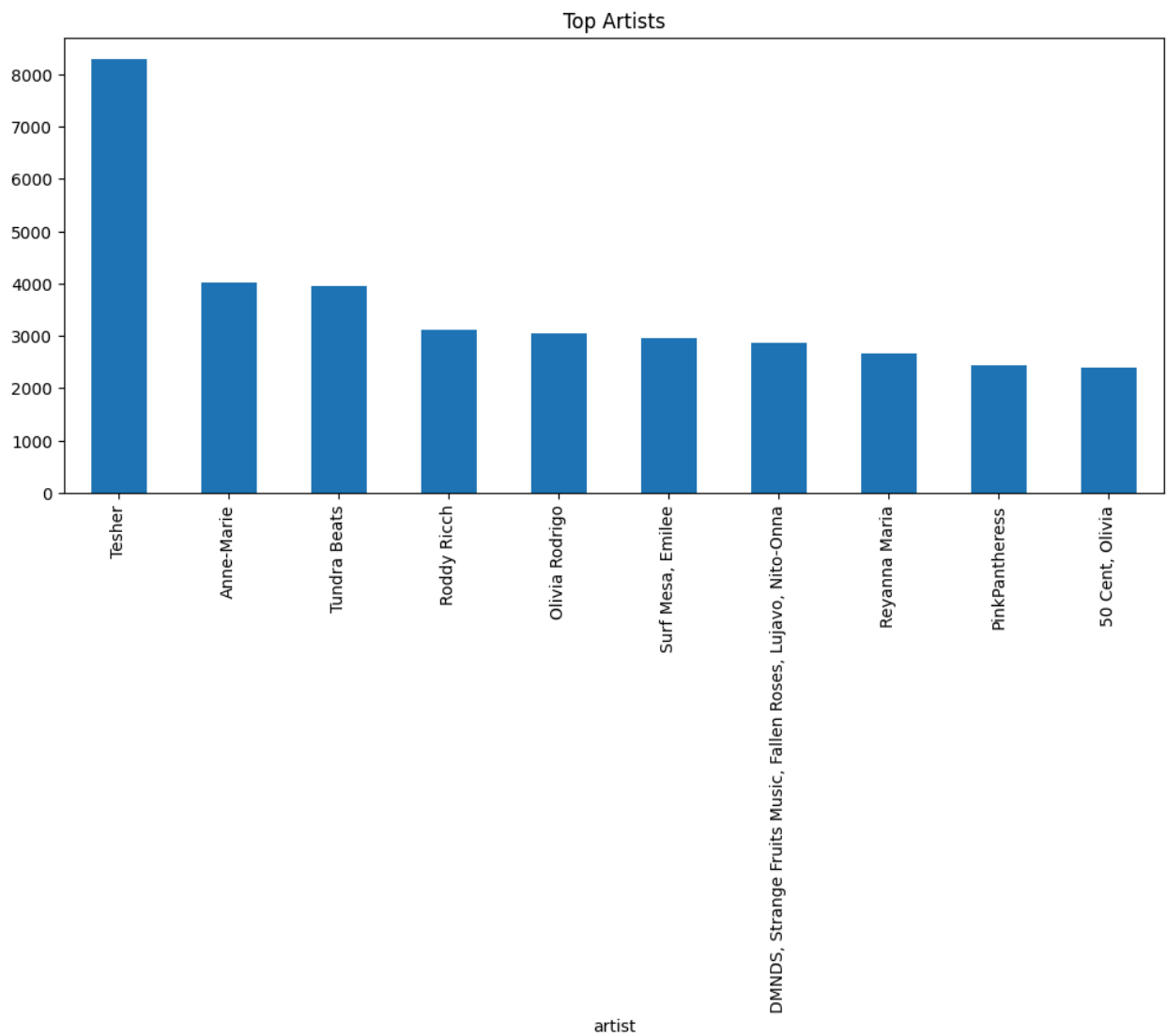
plt.figure(figsize=(12,5))

artist.plot(kind='bar')

plt.title("Top Artists")

plt.savefig("artists.png",dpi=300,bbox_inches="tight")

plt.show()
```



```
In [34]: #Top Albums

album=df['album'].value_counts().head(10)

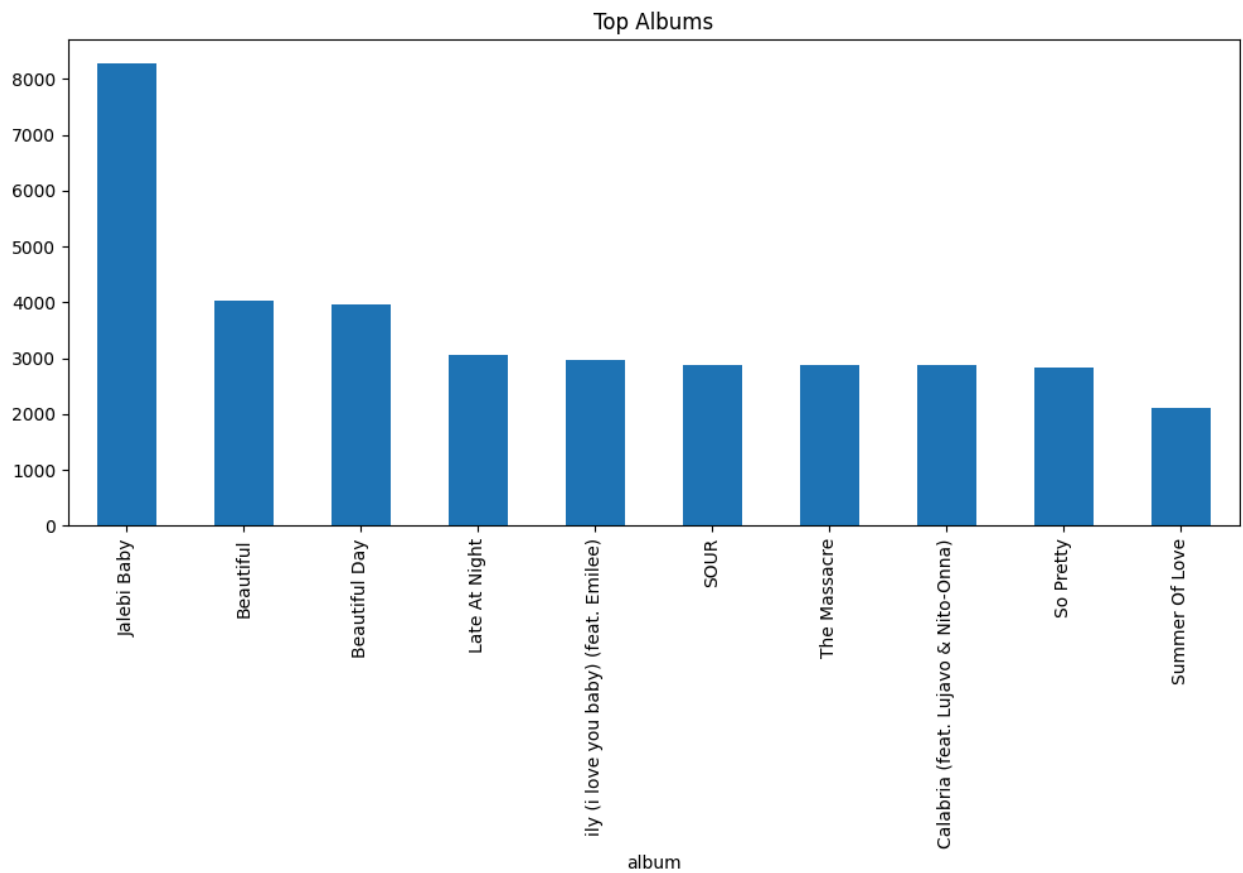
plt.figure(figsize=(12,5))

album.plot(kind='bar')

plt.title("Top Albums")

plt.savefig("albums.png",dpi=300,bbox_inches="tight")

plt.show()
```



```
In [35]: #Top Tracks

track=df['track'].value_counts().head(10)

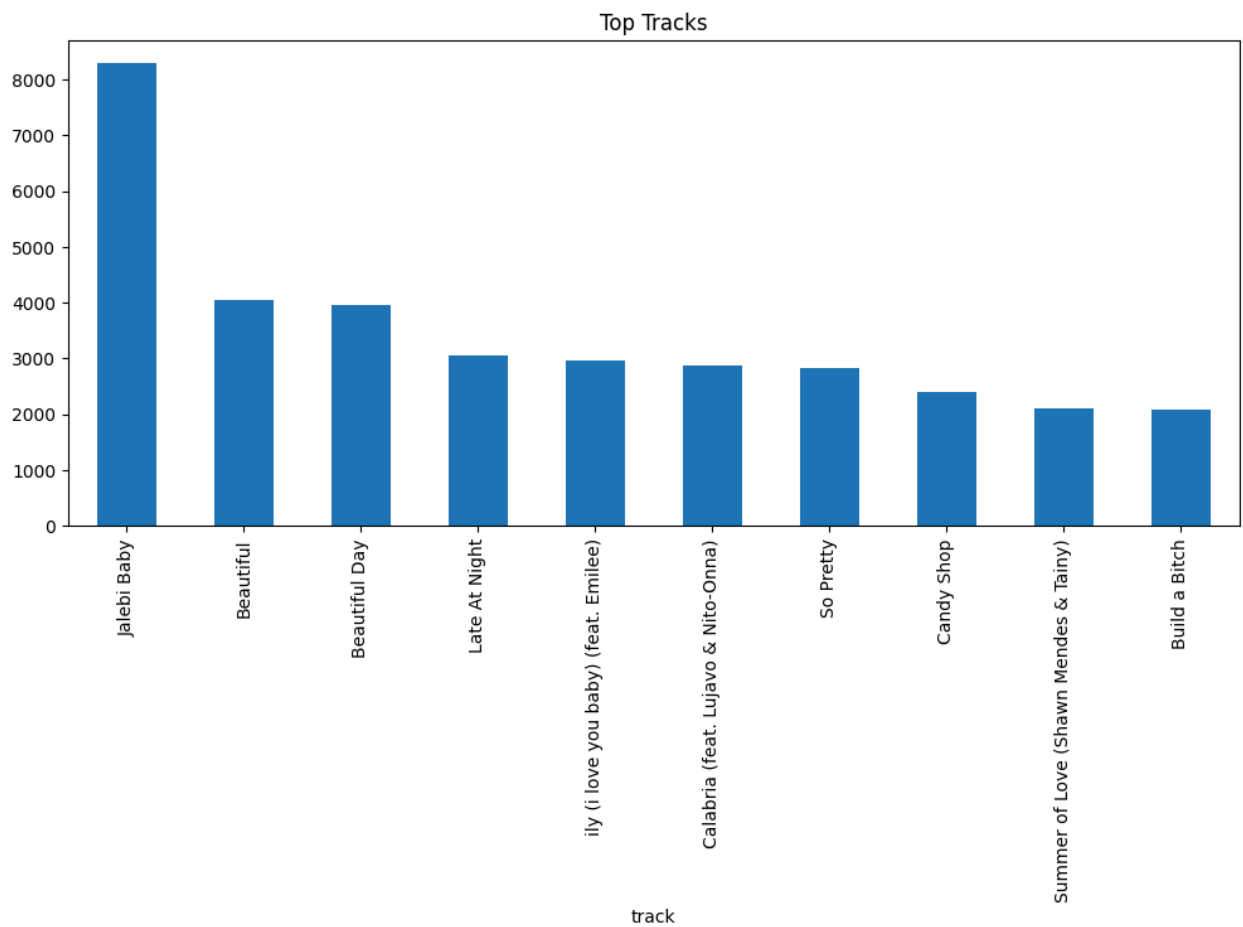
plt.figure(figsize=(12,5))

track.plot(kind='bar')

plt.title("Top Tracks")

plt.savefig("tracks.png",dpi=300,bbox_inches="tight")

plt.show()
```



```
In [36]: #Monthly Traffic

monthly=df.groupby('Month')['event'].count()

months=[
    'January', 'February', 'March', 'April', 'May', 'June',
    'July', 'August', 'September', 'October', 'November', 'December'
]

monthly=monthly.reindex(months)

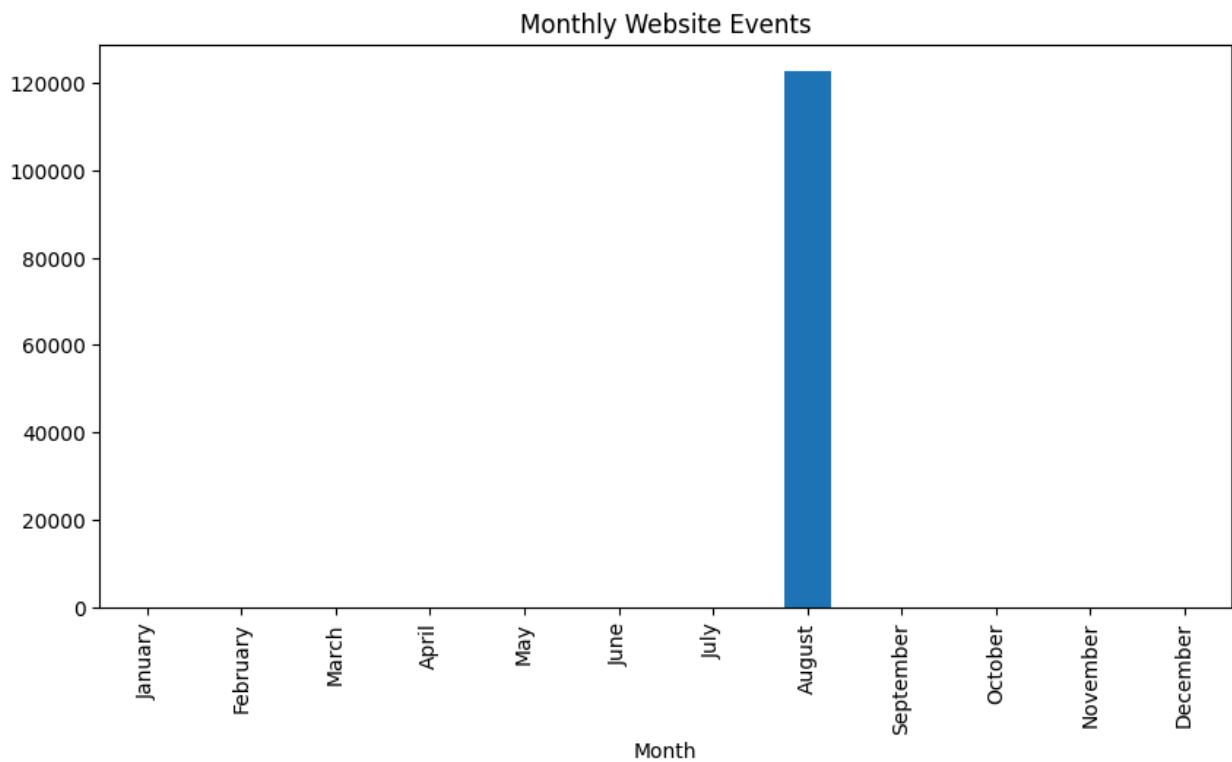
plt.figure(figsize=(10,5))

monthly.plot(kind='bar')

plt.title("Monthly Website Events")

plt.savefig("monthly.png",dpi=300,bbox_inches="tight")

plt.show()
```



```
In [37]: #Country Wise Event Type
pd.crosstab(df['country'],df['event']).head(10)
```

```
Out[37]:
```

	event	click	pageview	preview
--	-------	-------	----------	---------

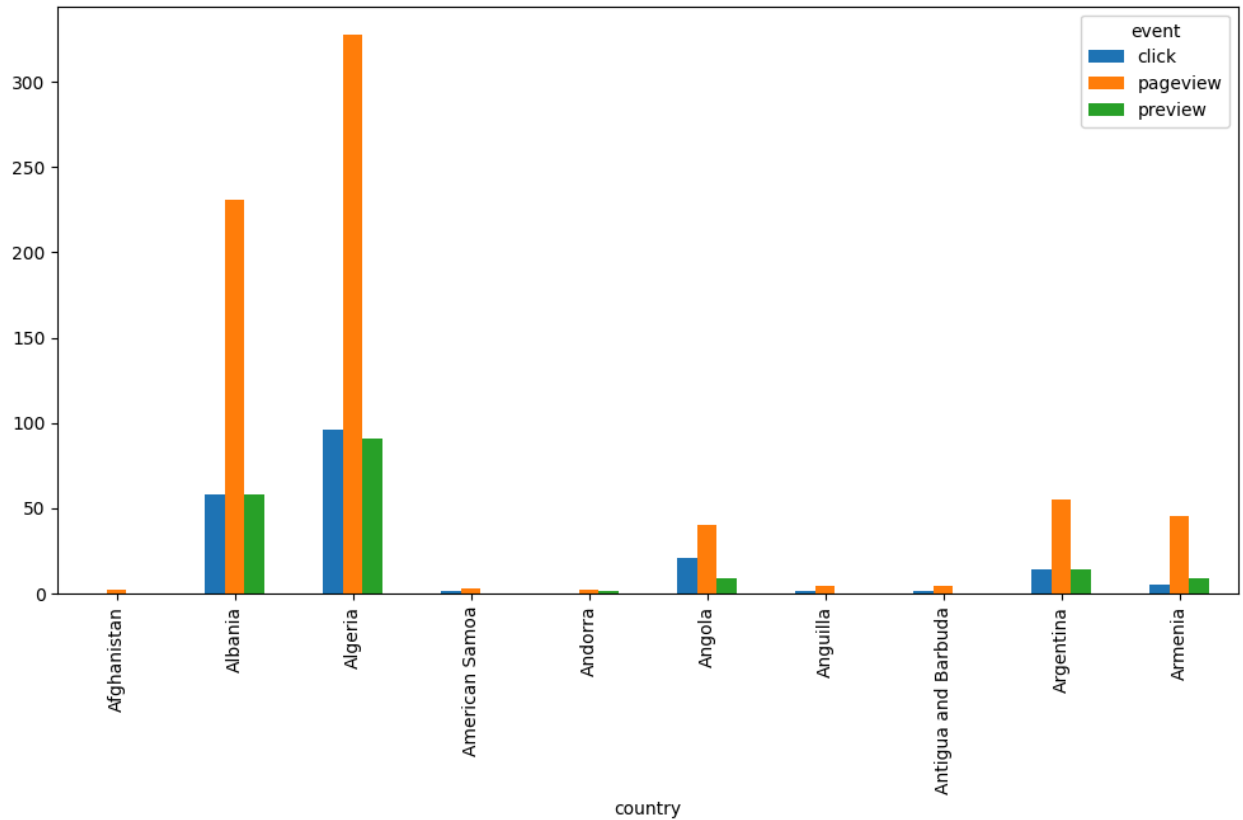
country				
Afghanistan	0	2	0	
Albania	58	231	58	
Algeria	96	328	91	
American Samoa	1	3	0	
Andorra	0	2	1	
Angola	21	40	9	
Anguilla	1	4	0	
Antigua and Barbuda	1	4	0	
Argentina	14	55	14	
Armenia	5	45	9	

```
In [38]: #visualization-country Wise Event Type

country_event=pd.crosstab(df['country'],df['event'])

country_event.head(10).plot(kind='bar',figsize=(12,6))
```

Out[38]: <Axes: xlabel='country'>



```
In [39]: print("Key Insights\n")  
  
print("1. Click events account for the majority of recorded website interactions.")  
print("2. Traffic is concentrated in a small number of countries and cities.")  
print("3. Certain artists, albums, and tracks receive significantly more user engagement than others.")  
print("4. Website activity remains relatively consistent over the available time period.")  
print("5. A small number of unique links generate most user interactions.")  
print("6. Geographic analysis helps identify high-engagement markets for targeted campaigns.")
```

Key Insights

1. Click events account for the majority of recorded website interactions.
2. Traffic is concentrated in a small number of countries and cities.
3. Certain artists, albums, and tracks receive significantly more user engagement than others.
4. Website activity remains relatively consistent over the available time period.
5. A small number of unique links generate most user interactions.
6. Geographic analysis helps identify high-engagement markets for targeted campaigns.

```
In [40]: print("Recommendations\n")
```

```
print("1. Focus marketing campaigns on high-traffic countries and cities.")  
print("2. Promote the most popular artists and tracks to maximize engagement.")  
print("3. Optimize underperforming links to improve click-through rates.")  
print("4. Analyze successful links and replicate their content strategy.")  
print("5. Expand promotional efforts in regions showing growing user activity.")
```

Recommendations

1. Focus marketing campaigns on high-traffic countries and cities.
2. Promote the most popular artists and tracks to maximize engagement.
3. Optimize underperforming links to improve click-through rates.
4. Analyze successful links and replicate their content strategy.
5. Expand promotional efforts in regions showing growing user activity.

Conclusion

This project analyzed website interaction data to gain insights into user engagement and content performance. The analysis identified the most common user events, top-performing countries, cities, artists, albums, and tracks, providing a better understanding of user behavior and geographic reach. These findings can help Alfido Tech focus marketing efforts on high-engagement regions, promote popular content, and improve overall user interaction.

It is important to note that the provided dataset does not include session identifiers, user identifiers, page URLs, or session duration. Therefore, metrics such as sessions, bounce rate, average session duration, landing pages, and exit pages could not be calculated. Despite these limitations, the event-level analysis provides meaningful insights that support data-driven business decisions and marketing optimization.